

Potential Impact Assessment

Minor AI for Society

Casey Stijlaart

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Introduction

This assessment evaluates the potential impacts of the AI-powered Smart Desk Plant Monitoring system. It considers the benefits, risks, and broader implications of the project for users, society, and the environment. The goal is to critically reflect on how the system could affect stakeholders, ensure responsible use of AI, and identify strategies to mitigate potential negative effects.

Project Overview

The proposed project is an AI-powered smart desk plant monitoring system that uses sensors and a TinyML model to monitor plant health. The system collects environmental data such as soil moisture, light intensity, temperature, and humidity. Based on patterns in this data, the AI model provides recommendations to improve plant care. A local dashboard allows users to view plant health indicators and historical trends.

The system is designed to run locally on an embedded device to ensure low latency and protect user privacy.

Stakeholders

The primary stakeholders of this system are individuals who keep plants at home or in office environments and want support in maintaining healthy plants. Secondary stakeholders may include hobby gardeners, technology enthusiasts, or educators who are interested in experimenting with AI-powered monitoring systems.

Indirect stakeholders could include individuals who benefit from improved environmental awareness or organizations exploring similar monitoring technologies for other applications such as indoor environmental monitoring.

Positive Impact

The project can have several positive impacts. First, it can help users better understand the environmental conditions that affect plant health, which may lead to improved plant care and reduced plant loss. This may also encourage users to develop a greater awareness of environmental factors such as light and humidity.

Second, the system demonstrates how AI can be applied in a practical and accessible way using embedded systems and TinyML. This can contribute to educational value by showcasing how AI can be implemented on small devices without requiring cloud infrastructure.

Finally, because the system processes data locally, it promotes privacy-friendly AI solutions that reduce dependence on external servers.

Risks and Negative Impacts

Despite its benefits, the system may also present potential risks. Sensor inaccuracies or limitations in the AI model could lead to incorrect recommendations, such as suggesting watering when it is not necessary. If users rely too heavily on the system, this could negatively affect plant care.

Another potential risk is that users may misunderstand the AI suggestions as definitive instructions rather than recommendations based on limited sensor data.

Additionally, the hardware required for the system, such as sensors and microcontrollers, introduces material and electronic resource use.

Privacy, Security, and Ethics

The project is designed with privacy in mind. All data processing takes place locally on the device, and no personal data is transmitted to external servers. This reduces privacy risks and aligns with privacy-by-design principles.

If a camera module is later added for plant classification, care should be taken to ensure that the system does not capture identifiable personal information or images of people.

From an ethical perspective, the system should present recommendations as suggestions rather than authoritative decisions, ensuring that the user remains in control.

Sustainability Considerations

Running AI models locally using TinyML reduces the need for cloud computing resources and continuous internet connectivity. This can reduce energy consumption associated with remote servers and data transmission.

However, the use of electronic components such as sensors, microcontrollers, and displays does contribute to material usage and electronic waste. Designing the system with durable components and low power consumption can help mitigate these effects.

Mitigation Strategies

Several strategies can help reduce potential risks. First, the system should clearly communicate that its recommendations are suggestions rather than guaranteed solutions. Second, calibration and validation of sensor data can help improve reliability.

In addition, designing the system to run locally and minimize unnecessary data collection helps protect user privacy. Using energy-efficient hardware and modular components can also improve sustainability and maintainability of the system.

Conclusion

The AI-powered Smart Desk Plant Monitoring system has the potential to provide valuable insights and support for plant care while demonstrating practical applications of AI on embedded devices. While there are potential risks related to sensor accuracy and user reliance, these can be mitigated through clear communication and careful design. Overall, the project promotes responsible AI use by prioritizing privacy and sustainability, while offering educational value and practical benefits for plant enthusiasts.